High-Resolution Bistatic Spotlight SAR Imagery With General Configuration and Accelerated Track

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***Abstract*— Due to the flexible configuration and maneuvering platform, bistatic synthetic aperture radar (BiSAR) plays an important role in modern remote sensing applications, but the nonideal track simultaneously introduces model mismatch and spatial-variant phase problems. This article proposes a subaperture (SA) parametric polar format algorithm (PFA) for high-resolution bistatic spotlight SAR (BSSAR) imaging. First, a bistatic parametric PFA (BPPFA) is proposed to focus on the generally configured BiSAR data. A high-precision range model in the bistatic range and ellipsoid parameter angle coordinate space is established to modify the PFA interpolation kernels. To further enhance the azimuth resolution, the PFA subimages are fused in the image domain based on the coordinate transformation between the unified Cartesian coordinates and local imaging polar coordinates. During the subimage fusion, in order to ensure the accurate projection and nonaliasing spectrum, we also consider the geometric deformation of the coarse PFA image and analyze the wavenumber support region along with the Nyquist sampling requirement. This novel SA method, which is theoretically more efficient than the fast back-projection algorithm (BPA), alleviates the limitation of full aperture resolution on PFA's depth of focus (DOF). Finally, our method is applied to the general BSSAR data, involving the level-flight airborne transmitter and the dive hypersonic vehicle-borne receiver, and the results of both points and distributed targets demonstrate its effectiveness and efficiency.**

***Index Terms*— Accelerated track, general bistatic configuration, polar format algorithm (PFA), subaperture (SA) technique, synthetic aperture radar (SAR).**

I. Introduction

BISTATIC spotlight synthetic aperture radar (BSSAR) imaging has broad application prospects in reconnaissance, anti-interference, and various other situations. Due to the flexibility of bistatic configuration, BSSAR also has many advantages over normal monostatic radar in terms of diverse scattering information acquisition [1], forward-looking imaging, lightweight payload, and so on [2]. However, the double-square-root (DSR) range equation makes the monostatic SAR imaging algorithms impractical for direct application [3]. Furthermore, in the high-frequency high-resolution radar system, the spatial-variant phase errors limit the depth of focus (DOF) in BSSAR imaging algorithms, as the integral aperture and motion deviations accumulate.

For BSSAR with complicated configuration, the spatial-variant characteristics of the radar echo come from three aspects: the bistatic configuration, the trajectory motion deviations, and the spotlight mode.

1) The most studied bistatic SAR (BiSAR) configurations in stripmap mode are categorized into two types: translational-invariant BiSAR and translational-variant BiSAR [4], [5], [6], [7], [8]. For translational-invariant BiSAR imaging, monostatic frequency-domain algorithms, such as the range-Doppler algorithm, chirp scaling algorithm, and omega-k algorithm, are modified by considering the additional range variance introduced by the difference between the monostatic and bistatic configurations [9], [10], [11], [12], [13]. Translational-variant BiSAR, including one stationary platform [14], [15], bistatic forward-looking [16], [17], and other nonparallel and nonequal-speed configurations, requires further compensation for the azimuth-variant phase errors [18], [19], [20], [21], [22]. The above two cases are usually restricted to the ideal uniform and straight trajectory.
2) Conventional processing, such as frequency-domain filtering, has limited ability to handle the spatial-variant errors caused by the real track's motion deviations. Based on the 2-D or range-Doppler spectrum, the frequency-domain algorithms derive the expression of the spatial-variant phase and compensate for it, whereas the bistatic configuration further increases their complexity, and these algorithms, which are dependent on the form of motion errors, are suitable for a limited number of bistatic cases. The study [23] is simplified to consider the range variance of the transmitter and the azimuth variance of the receiver under certain bounds, and [24] only verifies the small acceleration track. For thriving maneuverable platforms, such as hypersonic aircraft and reusable spacecraft, the spatial-variant echo

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phase is much more sensitive to the model approximation. Although it is possible to perform the universal but approximate strategy, such as image division and post-filtering in [25], the precise motion compensation for general BiSAR configuration is still a challenging task.

3) Spotlight SAR continuously illuminates the target area, and the slant range history is different for different targets, so the spatial-variant phase is an inherent characteristic of spotlight data. In this case, the back-projection algorithm (BPA) and the polar format algorithm (PFA) are usually used for focusing. However, the bistatic fast factorized BPA (BFFBPA) in [26] is still time-consuming due to the backgeocoded digital elevation model (DEM) of all the subimage grids involved in BFFBPA splitting stages. Meanwhile, the DOF of PFA is limited by residual interpolation errors, such as the wave-front curvature errors in the conventional PFA, which is further exacerbated by the drastic coupling and spatial-variant motion errors, as the aperture length increases. If high resolution is demanded, PFA faces a conflict between resolution and DOF.

There are kinds of attempts to solve the DSR range equation problem. Frequency-domain algorithms that rely on the azimuth-shift invariance have been developed with the accuracy improvements of the 2-D frequency spectrum or range-Doppler spectrum [27], [28], [29], [30]. Since wavenumber and space are a pair of Fourier transform (FT), one of the research directions of wavenumber-domain algorithms is to find an appropriate imaging space and the corresponding wavenumbers [31], [32], [33]. For the most general Euclidean slant range equation, the equivalent slant range model is first established by transforming the Cartesian coordinates into the imaging coordinates. Imaging coordinates are typically set according to the radar radiation geometry: [32] uses polar-range polar-angle imaging coordinates for monostatic configuration, and [33] uses series reversion approximation to derive the bistatic-range Doppler-aperture imaging coordinates for the stationary transmitter configuration. Compared with conventional PFA, they do not use Cartesian coordinates but geometry-dependent parametric coordinates for interpolation, which has shown that coordinate transformation combined with wavenumber interpolation makes PFA more effective and concise in extreme conditions, such as large squinted angles and more flexible maneuvers. Hence, the difficulty in solving the DSR range equation lies in the derivation of the accurate signal spectrum for frequency-domain algorithms; correspondingly, it could lie in the establishment of the accurate coordinate transformation for wavenumber-domain algorithms.

To address the above problems, this article proposes a subaperture bistatic parametric PFA (SA-BPPFA) to focus on high-resolution SAR imagery with a general bistatic configuration and complex radar track. Based on the equivalent DSR range equation and the bistatic configuration parameters, the relationship between imaging coordinates and Cartesian coordinates is built, and a BPPFA is proposed to focus on BSSAR data. To further improve the azimuth resolution, the SA technique is utilized, and the subimages are fused in the image domain through the projection from the local imaging plane to the unified Cartesian coordinate plane. At the same time, SA-BPPFA relaxes the restrictions of BPPFA on the DOF.

The main contributions of this article are as follows.

1) BPPFA can process the generally configured BSSAR data. The modified PFA in [32], [33], and [34] eliminates the planar wave-front assumption by converting the focusing plane for the specific configuration. However, for more general bistatic irradiation geometry, it is difficult to find proper imaging coordinates and ensure the accuracy of the coordinate transformation at the same time. In this article, a bistatic range–ellipsoid parametric angle image plane is proposed with the help of the bistatic ellipsoid geometry and its parameter equation, which lays no approximation in the coordinate transformation and, thus, is suitable for the general bistatic configuration.
2) As a kind of SA algorithm, SA-BPPFA not only improves the imaging resolution but also is suitable for nonideal radar tracks. There is a premise here that the PFA's wavenumbers are the functions of the radar motion parameters and reflect the azimuth spatial sampling of the radar echo. The uniform resampling of the PFA wavenumbers realizes the space conversion of SAR data, and during this process, the motion compensation in the traditional sense is also accomplished. Meanwhile, our coordinate-transform-based equivalent DSR range model accommodates precise radar motion parameters. Therefore, unlike the slant range model of the SA BPA [26], which relies on the ideal constant-speed and constant-height path, the SA fusion process with our novel slant range model is inherently applicable to complex trajectories.
3) The proposed BPPFA deduces the analytical expressions for the precise focus position after wavenumber interpolation. Also, we only need to calculate the backgeocoded DEM once to realize the projection of the subimages to the unified Cartesian grid. Compared with the bistatic SA BPA strategy in [26], SA-BPPFA both reduces the interpolation errors and improves the efficiency of backgeocoding.
4) In this article, we also derive the bistatic spectrum support region and a bistatic rotation angle (BRA) with the Cartesian coordinates. On this basis, the bistatic resolutions and the Nyquist sampling requirements using the two kinds of coordinates are given, which guides the design of the Cartesian grid size.

The rest of this article is organized as follows. In Section II, the geometry and range model for BSSAR are described; the imaging coordinate and the transformation relationship are derived based on the configuration parameters. In Section III, the proposed BPPFA and SA-BPPFA are introduced. In Section IV, the practical considerations are derived and discussed. In Section V, two simulated experiments and the analysis are performed to validate the effectiveness and

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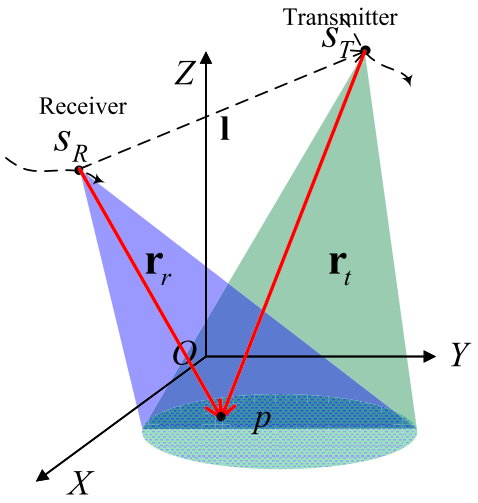


Fig. 1. Data acquisition geometry for BSSAR.

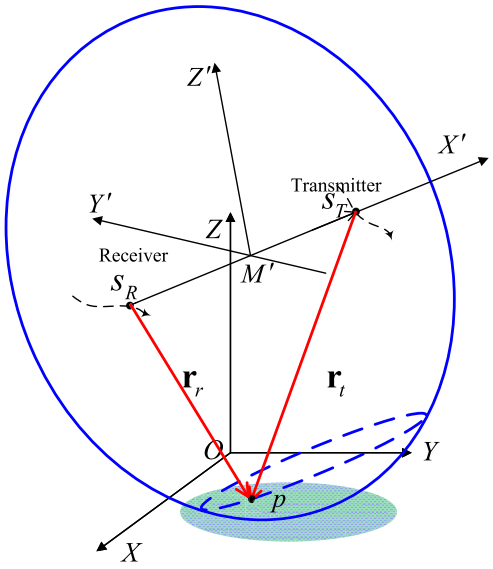


Fig. 2. Ellipsoid coordinate system for BSSAR.

efficiency improvement of SA-BPPFA. Finally, we present some conclusions in Section VI.

II. Geometry and Range Model

The data acquisition geometry for BSSAR is shown in Fig. 1. In the Cartesian coordinate system, the transmitter travels a curved track with the initial velocity $\mathbf{v}_t = [v_{tx}, v_{ty}, v_{tz}]^T$ and acceleration $\mathbf{a}_t = [a_{tx}(t_m), a_{ty}(t_m), a_{tz}(t_m)]^T$; the receiver is also mounted on a motion platform with initial velocity $\mathbf{v}_r = [v_{rx}, v_{ry}, v_{rz}]^T$ and acceleration $\mathbf{a}_r = [a_{rx}(t_m), a_{ry}(t_m), a_{rz}(t_m)]^T$, respectively. The high-order time-varying accelerations accurately describe the motion deviations, and it is noted that one platform could be stationary. At the synthetic aperture center moment t_0 (also the focus moment), the coordinates of the transmitter s_t, the receiver s_r, and a target point p in the scene are $[x_t, y_t, z_t]^T$, $[x_r, y_r, z_r]^T$, and $[x, y, z]^T$. The length of the baseline vector $\overrightarrow{s_r s_t}$ is l. At azimuth moment t_m, the DSR range equation for the target p is written as (1)–(3), shown at the bottom of the page, where $\|\cdot\|_2$ denotes the L^2-norm and

$$\begin{cases} \Delta_r(t_m) = \mathbf{v}_r t_m + \iint \mathbf{a}_r(t_m) dt_m dt_m \\ \Delta_t(t_m) = \mathbf{v}_t t_m + \iint \mathbf{a}_t(t_m) dt_m dt_m \end{cases} \tag{4}$$

represent the displacements of the transmitter and the receiver.

To find an appropriate imaging space, we first analyze the bistatic configuration and derive the coordinate transformation between the Cartesian and the imaging coordinates. The derivation mainly uses some spatial affine transformations and the ellipsoid parametric equations. As shown in Fig. 2, at the moment of azimuth focusing, the bistatic range of point target p is assumed to be ρ. p is on the ellipsoid with s_r and s_t as the focal points, and M' is the midpoint of the segment $\overline{s_r s_t}$, with coordinates as follows:

$$\begin{cases} x_M = (x_t + x_r)/2 \\ y_M = (y_t + y_r)/2 \\ z_M = (z_t + z_r)/2. \end{cases} \tag{5}$$

Another Cartesian coordinate system $M'X'Y'Z'$ is established in Fig. 2 with M' as the origin and $\overline{s_r s_t}$ as the X'-axis. The coordinates in $M'X'Y'Z'$ are shifted and rotated from $OXYZ$ with the relationship

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \mathbf{B}(\beta)\mathbf{A}(\alpha) \begin{bmatrix} x - x_M \\ y - y_M \\ z - z_M \end{bmatrix} \tag{6}$$

where $\mathbf{A}(\alpha)$ and $\mathbf{B}(\beta)$ are the rotation matrices that rotate α around the Z-axis and β around the Y'-axis, and they are expressed as follows ($\mathrm{sgn}(\cdot)$ is the sign function):

$$\begin{cases} \tan\alpha = (y_r - y_t)/(x_r - x_t) \\ \sin\beta = \mathrm{sgn}(x_t - x_r) \cdot (z_t - z_r)/l \end{cases} \tag{7}$$

$$\mathbf{A}(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{8}$$

$$R_r(t_m; x, y, z) = \|\mathbf{R}_r(t_m; x, y, z)\|_2 = \sqrt{[x_r + \Delta_{rx}(t_m) - x]^2 + [y_r + \Delta_{ry}(t_m) - y]^2 + [z_r + \Delta_{rz}(t_m) - z]^2} \tag{1}$$

$$R_t(t_m; x, y, z) = \|\mathbf{R}_t(t_m; x, y, z)\|_2 = \sqrt{[x_t + \Delta_{tx}(t_m) - x]^2 + [y_t + \Delta_{ty}(t_m) - y]^2 + [z_t + \Delta_{tz}(t_m) - z]^2} \tag{2}$$

$$R(t_m; x, y, z) = R_t(t_m; x, y, z) + R_r(t_m; x, y, z) \tag{3}$$

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$$\mathbf{B}(\beta)=\begin{bmatrix}\cos\beta & 0 & \sin\beta\\ 0 & 1 & 0\\ -\sin\beta & 0 & \cos\beta\end{bmatrix}. \tag{9}$$

The inverse of this process yields

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\mathbf{A}(-\alpha)\mathbf{B}(-\beta)\begin{bmatrix}x'\\ y'\\ z'\end{bmatrix}+\begin{bmatrix}x_M\\ y_M\\ z_M\end{bmatrix}. \tag{10}$$

In $M'X'Y'Z'$, one form of the standard ellipsoid parametric equation is (others are shown in Appendix A)

$$\begin{cases}x'=a\sin\theta\cos\varphi\\ y'=b\cos\theta\\ z'=b\sin\theta\sin\varphi\end{cases},\quad 0\leqslant\theta\leqslant\pi, 0\leqslant\varphi<2\pi \tag{11}$$

where $a=\rho/2$ and $b=(a^2-(l/2)^2)^{1/2}$; θ and φ are the ellipsoid parametric angles. According to the above derivations, the conversion relations between the Cartesian coordinates (x, y) and the ellipsoid parametric coordinates (ρ, φ) are derived below. With (6) and (11), the coordinate transformation from (x, y) to (ρ, φ) is shown in (12)–(15), at the bottom of the next page. With (10) and (11), the coordinate transformation from (ρ, φ) to (x, y) is shown in (16)–(18)

$$\begin{cases}x(\rho,\varphi)=x_M+a\cos\varphi\cos\alpha\cos\beta\sin[\theta(\rho,\varphi)]\\ \qquad\qquad -b\sin\alpha\cos[\theta(\rho,\varphi)]\\ \qquad\qquad -b\sin\varphi\cos\alpha\sin\beta\sin[\theta(\rho,\varphi)]\\ y(\rho,\varphi)=y_M+a\cos\varphi\sin\alpha\cos\beta\sin[\theta(\rho,\varphi)]\\ \qquad\qquad +b\cos\alpha\cos[\theta(\rho,\varphi)]\\ \qquad\qquad -b\sin\varphi\sin\alpha\sin\beta\sin[\theta(\rho,\varphi)]\end{cases} \tag{16}$$

where

$$\begin{cases}\sin[\theta(\rho,\varphi)]=(z-z_M)/(a\cos\varphi\sin\beta+b\sin\varphi\cos\beta)\\ \cos[\theta(\rho,\varphi)]=\xi_\theta\cdot\cos[\arcsin(\sin[\theta(\rho,\varphi)])]\end{cases} \tag{17}$$

and

$$\begin{cases}\xi_\theta=+1, & 0\leq\theta(\rho_c,\varphi_c)\leq\dfrac{\pi}{2}\\ \xi_\theta=-1, & \dfrac{\pi}{2}<\theta(\rho_c,\varphi_c)\leq\pi.\end{cases} \tag{18}$$

In (18), (ρ_c, φ_c) represents the reference target. The value of ξ_θ is spatial-variant referring to (14); to avoid the ambiguity of $\theta(\rho, \varphi)$, the algorithm requires that the whole observed scene must locate on the same half-Y'-axis region or change the form of the ellipsoid parametric equation.

In the above derivations, there is no approximation at all. Using operators $\mathbf{C}_p$ and $\mathbf{P}_c$ to denote the Cartesian-parametric coordinate transformation in (12) and the parametric-Cartesian transformation in (16), respectively, the coordinate transformations can be summarized as follows:

$$\begin{cases}(\rho,\varphi)=\mathbf{C}_p(x,y)\\ (x,y)=\mathbf{P}_c(\rho,\varphi).\end{cases} \tag{19}$$

Substituting (16) into (1)–(3), the equivalent DSR range in bistatic range–ellipsoid parameter angle space is rewritten in (20)–(22), as shown at the bottom of the next page

$$R(t_m;\rho,\varphi,z)=R_t(t_m;\rho,\varphi,z)+R_r(t_m;\rho,\varphi,z). \tag{22}$$

III. Subaperture BPPFA

In this section, we first introduce the BPPFA implemented in the SAs for coarse focusing and analyze its performance; then, the wavenumber support region and Nyquist sampling requirements for both coarse focusing and SA fusion are derived; finally, we conclude the processing steps and provide the complete implementation flow for SA-BPPFA.

A. Bistatic Parametric PFA

Given that the transmitted signal is a linear frequency modulated (LFM) with pulsewidth T_p and chirp rate γ, the demodulated echo is

$$s(t,t_m;x,y,z)=\mathrm{rect}\left(\frac{t-t_p}{T_p}\right)\cdot\exp\left[-j2\pi f_c t_p+j\pi\gamma\cdot(t-t_p)^2\right] \tag{23}$$

where f_c is the carrier frequency, $t_p=R(t_m;x,y,z)/c$ is the fast time delay, $t_m\in[-(T_a/2)+t_0,(T_a/2)+t_0)$ is the slow time, and T_a is the azimuth synthetic duration.

After the fast FT (FFT) with respect to the fast time, range compression, and bulk compression, the signal turns into [34]

$$s_{bc}(f_r,t_m;x,y,z)=\exp\left\{-j\frac{2\pi(f_r+f_c)}{c}\left[R(t_m;x,y,z)-R(t_m;x_c,y_c,z_c)\right]\right\} \tag{24}$$

where f_r is the range sampling frequency. The reference target is the scene center point, and its Cartesian coordinate is (x_c, y_c, z_c). The bulk compression compensates for all the spatial-invariant terms, including the error terms caused by the track deviations [35]. Then, the Cartesian coordinates are replaced by the imaging coordinates, and the signal is rewritten as follows:

$$s_{bc}(f_r,t_m;\rho,\varphi,z)=\exp\left\{-j\frac{2\pi(f_r+f_c)}{c}[R(t_m;\rho,\varphi,z)-R(t_m;\rho_c,\varphi_c,z_c)]\right\}. \tag{25}$$

The Taylor series of the differential range is shown as follows:

$$R(t_m;\rho,\varphi,z)-R(t_m;\rho_c,\varphi_c,z_c)=k_0(\rho,\varphi,z)+k_1(\rho,\varphi,z)\cdot(t_m-t_0)+R_n(t_m-t_0) \tag{26}$$

where $R_n(\cdot)$ represents the residual term. Obviously, $k_0(\rho,\varphi,z)=\rho-\rho_c$. Then, the keystone transform is performed by resampling the slow time for each range frequency as follows:

$$(t_m-t_0)\rightarrow\left(\frac{f_c}{f_c+f_r}\right)\cdot(t_m-t_0) \tag{27}$$

and the signal turns into (28), as shown at the bottom of the next page. Usually, the range cell migration can be approximated as its first-order Taylor series in short apertures to achieve 2-D decoupling. After the range inverse FT (IFT), the decoupled signal focused at the range position ρ is shown as follows:

$$s_{de}(\rho_s,t_m;\rho,\varphi,z)$$

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$$\approx \operatorname{sinc}\left(\frac{\rho_s-\rho}{\Delta_\rho}\right) \cdot \exp\{-jk_{rc}\cdot[R(t_m;\rho,\varphi,z)-R(t_m;\rho_c,\varphi_c,z_c)]\} \quad (29)$$

where $k_{rc}=2\pi f_c/c$, Δ_ρ is the range resolution, and ρ_s is the sampling range.

In the azimuth processing, the DSR range $R(t_m;\rho,\varphi,z)$ will first be expanded into the first-order Taylor approximation at $\varphi=\varphi_c$, which is given by

$$R(t_m;\rho,\varphi,z) = R(t_m;\rho,\varphi_c,z_c)+\kappa(t_m;\rho,z_c)\cdot(\varphi-\varphi_c)+R_n(\varphi-\varphi_c) \quad (30)$$

where

$$\kappa(t_m;\rho,z_c)=\left.\frac{\partial R(t_m;\rho,\varphi,z_c)}{\partial\varphi}\right|_{\varphi=\varphi_c} \qquad t_m\in\left[-\frac{T_a}{2}+t_0,\frac{T_a}{2}+t_0\right) \quad (31)$$

is the function of t_m, and it is called the scaling azimuth time. The range-dependent slant range $R(t_m;\rho,\varphi_c,z_c)$ in (30) is independent of the azimuth coordinate, and we compensate for it first with (32), as shown at the bottom of the next page; then, the signal phase in (29) is rewritten in (33), as shown at the bottom of the next page.

This range-dependent operation, which operates in each range gate, compensates for the range-variant phase errors included in $R(t_m;\rho,\varphi_c,z_c)$. However, $R(t_m;\rho_s,\varphi_c,z_c)$ varies with both the azimuth slow time and the sampling range, and this operation is equivalent to multiplying different range sampling sequences for different azimuth pulses, which causes the range spectrum shifts, so the problem of range-variant motion compensation is transferred to the burden of range oversampling. As for $R(t_0;\rho_s,\varphi_c,z_c)$ in (33), its value is ρ_s, and it ensures that the range spectrum shift at t_0 is zero to avoid the spectrum aliasing. This slant range at the aperture center moment also distinguishes the radar position, and it is preserved for beamforming in the SA fusion.

With (30)–(33), we acquire the original azimuth wavenumber $k_\varphi = k_{rc}\kappa(t_m;\rho,z_c)$ and then rearrange phase $\Phi_\Delta(t_m;\rho,\varphi)$ by the scaling azimuth time to get (34), as shown at the bottom of the next page. Next, $\kappa(t_m;\rho,z_c)$ is interpolated to the uniform sampling form $\kappa_s=g_s t_m$, and the phase turns into

$$\Phi_2(\rho_s,\kappa_s;\rho,\varphi,z) = -k_{rc}\cdot\left[\rho_s+\psi(\rho,\varphi,z)\cdot\kappa_s+R_n(\kappa_s)+\tilde{R}(\kappa_s,\rho_s)\right] \quad (35)$$

where

$$\tilde{R}(\kappa_s,\rho_s)=R(\kappa_s;\rho,\varphi_c,z_c)-R(\kappa_s;\rho_s,\varphi_c,z_c) \quad (36)$$

and $\tilde{R}(\kappa_s,\rho)=0$ in range gate $\rho_s=\rho$, so this term does not affect the azimuth phase. Omitting the high-order term in (35), the signal in the range-space azimuth-wavenumber domain is shown as follows:

$$s_{\text{inter}}(\rho_s,\kappa_s;\rho,\varphi,z) = \varpi(\rho)\cdot\operatorname{sinc}\left(\frac{\rho_s-\rho}{\Delta_\rho}\right)\cdot\exp\left\{-jk_{rc}\left[\rho_s+\kappa_s\cdot\psi(\rho,\varphi,z)+\tilde{R}(\kappa_s,\rho_s)\right]\right\}. \quad (37)$$

$$\begin{cases}\rho(x,y)=\sqrt{(x_t-x)^2+(y_t-y)^2+(z_t-z)^2}+\sqrt{(x_r-x)^2+(y_r-y)^2+(z_r-z)^2}\\ \varphi(x,y)=\xi_{\varphi_1}+\xi_{\varphi_2}\cdot\arcsin\left\{\left[-(x-x_M)\cos\alpha\sin\beta-(y-y_M)\sin\alpha\sin\beta-(z-z_M)\cos\beta\right]/b/\sin\left[\theta(x,y)\right]\right\}\end{cases} \quad (12)$$

where

$$\begin{cases}\sin\left[\varphi(x,y)\right]=\dfrac{-(x-x_M)\cos\alpha\sin\beta-(y-y_M)\sin\alpha\sin\beta+(z-z_M)\cos\beta}{b\sin\left[\theta(x,y)\right]}\\ \cos\left[\varphi(x,y)\right]=\dfrac{(x-x_M)\cos\alpha\cos\beta+(y-y_M)\sin\alpha\cos\beta+(z-z_M)\sin\beta}{a\sin\left[\theta(x,y)\right]}\end{cases} \quad (13)$$

$$\theta(x,y)=\arccos\left[\frac{-(x-x_M)\sin\alpha+(y-y_M)\cos\alpha}{b}\right] \quad (14)$$

and

$$\begin{cases}\xi_{\varphi_1}=\pi\ \&\ \xi_{\varphi_2}=-1, & \cos\left[\varphi(x,y)\right]<0\\ \xi_{\varphi_1}=0\ \&\ \xi_{\varphi_2}=+1, & \cos\left[\varphi(x,y)\right]\geq 0\ \&\ \sin\left[\varphi(x,y)\right]\geq 0\\ \xi_{\varphi_1}=2\pi\ \&\ \xi_{\varphi_2}=+1, & \cos\left[\varphi(x,y)\right]\geq 0\ \&\ \sin\left[\varphi(x,y)\right]<0\end{cases} \quad (15)$$

$$R_r(t_m;\rho,\varphi,z)=\sqrt{[x_r+\Delta_{rx}(t_m)-x(\rho,\varphi)]^2+\left[y_r+\Delta_{ry}(t_m)-y(\rho,\varphi)\right]^2+\left[z_r+\Delta_{rz}(t_m)-z\right]^2} \quad (20)$$

$$R_t(t_m;\rho,\varphi,z)=\sqrt{[x_t+\Delta_{tx}(t_m)-x(\rho,\varphi)]^2+\left[y_t+\Delta_{ty}(t_m)-y(\rho,\varphi)\right]^2+\left[z_t+\Delta_{tz}(t_m)-z\right]^2} \quad (21)$$

$$s_{de}(f_r,t_m;\rho,\varphi,z)=\exp\left\{-jk_{ra}\left[\rho-\rho_c+k_1(\rho,\varphi,z)\cdot\frac{f_c}{f_c+f_r}\cdot(t_m-t_0)+R_n\left(\frac{f_c}{f_c+f_r}(t_m-t_0)\right)\right]\right\} \quad (28)$$

where $k_{ra}=2\pi(f_r+f_c)/c$.

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The signal phase is rearranged in the form of the azimuth sampling wavenumber $k_{\varphi s} = k_{rc}\kappa_s$, which realized the transformation of the sampling domain. Due to the range dependence of the azimuth wavenumber, the azimuth wavenumber spectrum width changes, as the range increases. To ensure the conservation of energy, the amplitude of the wavenumber spectrum needs to be multiplied by the scaling factor $\varpi(\rho) = B_{\kappa_s}/B_{\kappa(t_m;\rho_s,z_c)}$, which is just the oversampling ratio for each range gate. Then, the focus image in (38), as shown at the bottom of the next page, is acquired after azimuth FFT.

The signal space and image space in the SAR echo are coupled with each other, and imaging algorithms realize the space conversion of SAR data [36]. The frequency represents the change rate of phase in the time domain, and the wavenumber represents the rate of phase change in the spatial domain. Thus wavenumber space is a pair of FT, and wavenumber-domain algorithms extract the 2-D imaging coordinates from the slant range equation to obtain the corresponding 2-D wavenumbers. Spatial conversion of echo data in most PFA [31], [32], [33], [34] is achieved through orthogonal decomposition and resampling of the 2-D wavenumbers. Orthogonal wavenumber decomposition can also be divided into two steps: first decoupling, and then, parametric PFA can compensate for the range-variant phase errors in the range-space domain [34]. In addition to PFA interpolation [32], the resampling of the 2-D wavenumbers is also classified as regularization or homogenization of nonuniform sampling in [33] and [34]. Since the wavenumber functions include all radar motion parameters and reflect the radar track, the accurate real-track range equation stimulates the inherent motion compensation ability of interpolation algorithms, and the coherent accumulation of energy is achieved by FFT after interpolating the data into the new wavenumber sampling domain.

Next, we analyze the performance of BPPFA, including its geometry deformation and the DOF.

The geometric deformation of BPPFA is caused by the difference between $\psi(\rho,\varphi,z)$ and the preset φ. According to (33) and (34), the final focus position is calculated as follows:

$$\psi(\rho,\varphi,z) = \left.\frac{\partial[R(t_m;\rho,\varphi,z) - R(t_m;\rho,\varphi_c,z_c)]}{\partial\kappa(t_m;\rho,z_c)}\right|_{\kappa(t_m;\rho,z_c)=\kappa(t_0;\rho,z_c)}. \tag{39}$$

By using the chain rule, (39) is rewritten as follows:

$$\psi(\rho,\varphi,z) = \left.\frac{\partial[R(t_m;\rho,\varphi,z) - R(t_m;\rho,\varphi_c,z_c)]}{g(\rho)\partial t_m}\right|_{t_m=t_0}. \tag{40}$$

After replacing (ρ,φ) with (x,y) in (40), we establish the relationship from the Cartesian coordinate to the final focus position of BPPFA and yield a new coordinate transformation as follows:

$$(\rho,\psi) = \mathbf{C}_c(x,y,z). \tag{41}$$

The final DOF of our proposed algorithm depends on the coarse focus images, and the DOF of BPPFA is influenced by two aspects; one is the residual range cell migration after decoupling, and the other is the residual azimuth phase errors after the first-order wavenumber interpolation. The phase errors limitation is much stricter, and the residual error is usually approximated by its lowest order term; then, the negligible case of the residual phase errors in (35) is

$$\frac{1}{2}|Q(\rho,\varphi,z)|\cdot\left(\frac{\gamma_a T_a}{2}\right)^2 \le \frac{\pi}{4} \tag{42}$$

where $Q(\rho,\varphi,z)$ is the second-order Taylor coefficient of $R_n[\kappa(t_m;\rho,z_c)]$ in (34). It is calculated as follows:

$$Q(\rho,\varphi,z) = \frac{1}{2}\left.\frac{\partial^2[R(t_m;\rho,\varphi) - R(t_m;\rho_s,\varphi_c)]}{\partial[\kappa(t_m;\rho,z_c)]^2}\right|_{\kappa(t_m;\rho,z_c)=\kappa(t_0;\rho_s,z_c)}. \tag{43}$$

Apart from the bistatic configuration and the radar motion parameters, the values of $Q(\rho,\varphi,z)$ are influenced by the 3-D target positions, which indicates the spatial-variant characteristic of SAR echo, and thus, the phase errors limit the effective DOF. The constraint in (42) also shows that the azimuth DOF is inversely proportional to the synthetic duration.

B. Wavenumber Support Region and Nyquist Sampling Requirements

In this section, the wavenumber support region and Nyquist sampling requirements for both the parametric coordinate used in the coarse focus processing and the Cartesian coordinate used in the SA fusion are analyzed.

1) For the Parametric Coordinate: Based on the coordinate transformation (19) and the compound function derivation law, the scaling azimuth time is (details in Appendix B)

$$\begin{aligned}\kappa(t_m;\rho,z_c) &= \left.\frac{\partial R(t_m;\rho,\varphi,z_c)}{\partial\varphi}\right|_{\varphi=\varphi_c}\\ &= \left.\frac{\partial R(t_m;x,y,z_c)}{\partial x}\cdot\frac{\partial x}{\partial\varphi} + \frac{\partial R(t_m;x,y,z_c)}{\partial y}\cdot\frac{\partial y}{\partial\varphi}\right|_{\varphi=\varphi_c}\\ &= f_x\cdot n_x + f_y\cdot n_y\big|_{\varphi=\varphi_c}.\end{aligned} \tag{44}$$

$$h_{rmoco}(t_m;\rho_s) = \exp\{-jk_{rc}\cdot\{R(t_m;\rho_c,\varphi_c,z_c) - [R(t_m;\rho_s,\varphi_c,z_c) - R(t_0;\rho_s,\varphi_c,z_c)]\}\} \tag{32}$$

$$\Phi_1(\rho_s,t_m;\rho,\varphi,z) = -k_{rc}\cdot\{R(t_m;\rho,\varphi,z) - [R(t_m;\rho_s,\varphi_c,z_c) - R(t_0;\rho_s,\varphi_c,z_c)]\} \tag{33}$$

$$\Phi_1[\rho_s,\kappa(t_m;\rho,z_c);\rho,\varphi,z] = -k_{rc}\cdot\left\{\begin{array}{c}\rho_s + \psi(\rho,\varphi,z)\cdot\kappa(t_m;\rho,z_c) + R_n[\kappa(t_m;\rho,z_c)]\\ +R[\kappa(t_m;\rho,z_c);\rho,\varphi_c,z_c] - R[\kappa(t_m;\rho,z_c);\rho_s,\varphi_c,z_c]\end{array}\right\} \tag{34}$$

where $\psi(\rho,\varphi,z)$ is the final azimuth focus position.

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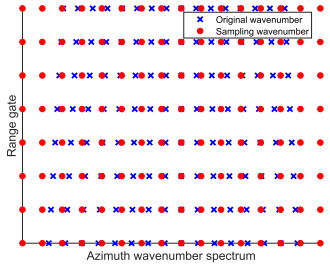


Fig. 3. Diagram for azimuth wavenumber interpolation.

In the same way, the derivative of the scaling azimuth time is

$$\begin{aligned} g(\rho) &= \left.\frac{\partial \kappa(t_m;\rho,z_c)}{\partial t_m}\right|_{t_m=t_0} \\ &= \left.\frac{\partial\left[f_x n_x + f_y n_y\right]}{\partial t_m}\right|_{t_m=t_0,\varphi=\varphi_c} \\ &= g_r(t_m;\rho,\varphi) + g_t(t_m;\rho,\varphi)|_{t_m=t_0,\varphi=\varphi_c}. \end{aligned} \tag{45}$$

The azimuth spectrum width of the radar echo is determined according to the value range of the azimuth wavenumber k_φ. Since the instantaneous derivative of $\kappa(t_m;\rho,z_c)$ changes negligibly within the short aperture, an alternative representation of its derivative is given here, which better explains the origin of the azimuthal wavenumber spectrum. According to the compound function derivation law, (45) can be expressed as follows:

$$\begin{aligned} g(\rho) &= \left.\frac{\partial \kappa(t_m;\rho,z_c)}{\partial t_m}\right|_{t_m=t_0} \\ &= -\lambda \left.\frac{\partial f_{dc}(\rho,\varphi,z_c)}{\partial \varphi}\right|_{\varphi=\varphi_c} \\ &= -\lambda \gamma_\varphi(\rho,\varphi_c,z_c). \end{aligned} \tag{46}$$

In (46), $\gamma_\varphi(\rho,\varphi_c,z_c)$ is the instantaneous Doppler changing rate projecting at position (ρ,φ_c,z_c). Then, similar to the Doppler bandwidth, the azimuthal wavenumber spectral width is proportional to the aperture synthetic time.

The requirement for the azimuth sampling wavenumber is given as follows:

$$g_s = \varpi_m \cdot \max_{\rho,t_0}\left[g(\rho)\right]. \tag{47}$$

$k_{\varphi s} = k_{rc}\kappa_s$ is the azimuth sampling wavenumber, $\Delta k_{\varphi s} = k_{rc} g_s \mathrm{PRF}$ is the azimuth sampling interval (PRF is the pulse repetition frequency), and ϖ_m is the oversampling rate that satisfies the Nyquist sampling theorem. As shown in (31) and Fig. 3, $\kappa(t_m;\rho,z_c)$ is range-dependent, so not only g_s should be greater than the maximum of $g(\rho)$, which guarantees the spectrum utilization and continuity, but also it is better to ensure the sampling interval is the same for all range gates to simplify the SA fusion.

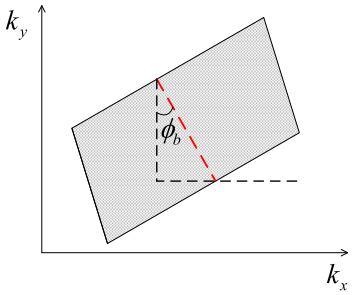


Fig. 4. Spectrum rotation angle.

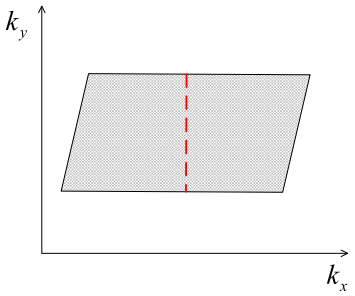


Fig. 5. Rotated spectrum.

TABLE I
RADAR SYSTEM PARAMETERS

Bandwidth	100 MHz
Carrier Frequency	9.6 GHz
Pulse Repetition Frequency	2000 Hz
Synthetic Time	1.024 s

By the way, the azimuth wavenumber change rate at t_0 is

$$\gamma_a = k_{rc} g(\rho) \tag{48}$$

and the azimuth resolution is

$$\Delta_\varphi = \frac{2\pi}{|\gamma_a T_a|}. \tag{49}$$

As for the range spectrum, the range IFT of (38) yields the new range wavenumber as follows:

$$\tilde{k}_{ra} = \frac{2\pi f_r}{c} + k_{rc}\cdot[1+\varepsilon(\kappa_s)] \tag{50}$$

where

$$\varepsilon(\kappa_s) = \frac{\partial \tilde{R}(\kappa_s,\rho_s)}{\partial \rho_s}. \tag{51}$$

In (51), $\varepsilon(\kappa_s)$ shifts with κ_s, which means that the range spectrum is shifted with azimuthal wavenumber samples, and the whole range bandwidth needs to be oversampled to avoid range ambiguity.

2) For the Cartesian Coordinate: Inspired by the conventional polar coordinate algorithm, a general calculation method of the Cartesian coordinate wavenumber support region is presented.

$$I(\rho_s,\varphi_s;\rho,\varphi) = \varpi(\rho)\mathrm{sinc}\left(\frac{\rho_s-\rho}{\Delta_\rho}\right)\sin c\left[\frac{\varphi_s-\psi(\rho,\varphi,z)}{\Delta_\varphi}\right]\exp\left\{-jk_{rc}\left[\rho_s+\tilde{R}(\varphi_s,\rho_s)\right]\right\} \tag{38}$$

where Δ_φ is the azimuth resolution and φ_s is the azimuth sampling angle.

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TABLE II
CONFIGURATION PARAMETERS BEFORE ROTATION

Configuration	Transmitter Location (km)	Receiver Location (km)	Transmitter Velocity (m/s)	Receiver Velocity (m/s)	BRA (rad)
I	(-5,-4,2)	(-10,-30,20)	(100,100,0)	(0,1700,-50)	-0.63
II	(-2,-3,2)	(-3,-5,10)	(200,0,0)	(200,0,0)	-0.57
III	(-2,-3,2)	(-3,-5,10)	(200,0,0)	(0,0,0)	-0.57

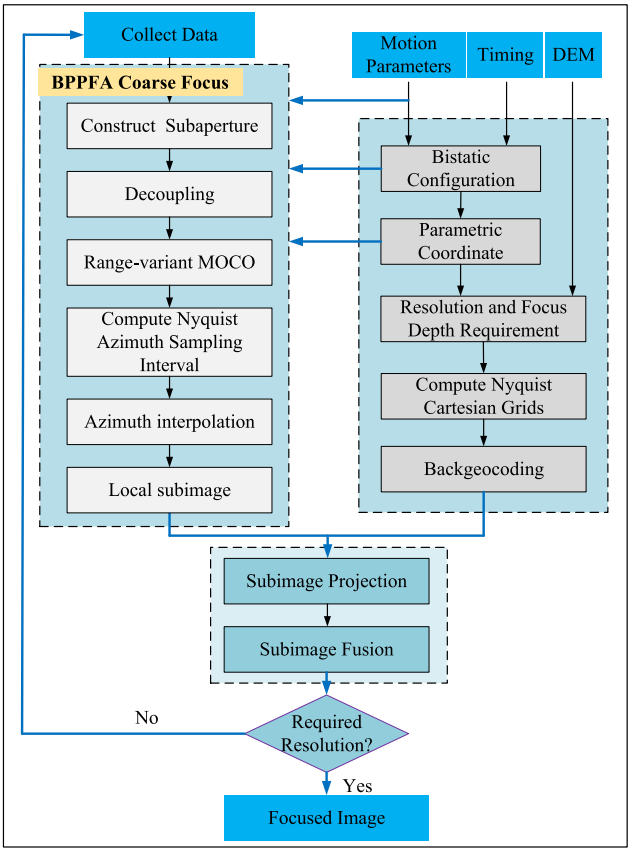


Fig. 6. Flowchart of the proposed algorithm.

When the instantaneous range is expanded to the first-order term of 2-D Cartesian coordinates, large position errors remain. Therefore, we first expand the instantaneous range to the fourth-order term of the slow time, and then further expand these coefficients to the first-order term of the Cartesian coordinates to obtain the more precise 2-D Cartesian wavenumber expression [37]; the derivations are shown as follows:

$$R(t_m; x, y, z) \approx \sum_{n=0}^{4} \zeta_n(x, y, z) \cdot (t_m - t_0)^n \tag{52}$$

$$\zeta_n(x, y, z) \approx \zeta_n(x_0, y_0, z_0) + \varsigma_{nx} \cdot (x - x_0) + \varsigma_{ny} \cdot (y - y_0) \tag{53}$$

where

$$\begin{cases} \varsigma_{nx} = \left.\dfrac{\partial \zeta_n(x, y, z)}{\partial x}\right|_{(x,y,z)=(x_0,y_0,z_0)} \\ \varsigma_{ny} = \left.\dfrac{\partial \zeta_n(x, y, z)}{\partial y}\right|_{(x,y,z)=(x_0,y_0,z_0)} \end{cases}. \tag{54}$$

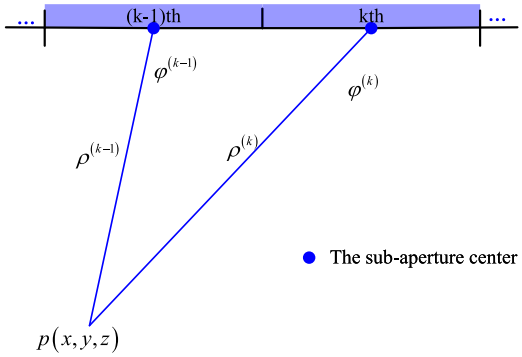


Fig. 7. Schematic of the relationship between Cartesian coordinates and local SA polar coordinates.

Substitute (53) into (52) to get

$$\begin{aligned} R(t_m; x, y, z) \approx & \sum_{n=0}^{4} \zeta_n(x_0, y_0, z) \cdot (t_m - t_0)^n \\ & + \sum_{n=0}^{4} \varsigma_{nx} (t_m - t_0)^n \cdot (x - x_0) \\ & + \sum_{n=0}^{4} \varsigma_{ny} (t_m - t_0)^n \cdot (y - y_0) \end{aligned} \tag{55}$$

and the 2-D wavenumbers are

$$\begin{cases} k_x(k_{ra}, t_m) = k_{ra} \cdot \sum_{n=0}^{4} \varsigma_{nx} (t_m - t_0)^n \\ k_y(k_{ra}, t_m) = k_{ra} \cdot \sum_{n=0}^{4} \varsigma_{ny} (t_m - t_0)^n. \end{cases} \tag{56}$$

Substitute $(x_0, y_0, z_0) = (x, y, z)$ in (56), and the wavenumbers at (x, y, z) are acquired. Since the wavenumber support region is spatial-variant, the Nyquist sampling requirement requires the sampling wavenumbers to cover the wavenumber support region of all targets, which is expressed by the following equations:

$$\begin{cases} k_{xs} \in \omega_x \cdot \left[\min\limits_{(x,y,z)} (k_x), \max\limits_{(x,y,z)} (k_x) \right] \\ k_{ys} \in \omega_y \cdot \left[\min\limits_{(x,y,z)} (k_y), \max\limits_{(x,y,z)} (k_y) \right] \end{cases} \tag{57}$$

where k_{xs} and k_{ys} are the sampling wavenumbers of 2-D Cartesian coordinates; ω_x and ω_y are the oversampling ratios. The Cartesian coordinate sampling grid sizes are

$$\begin{cases} \Delta_{xs} = \dfrac{2\pi}{B(k_{xs})} \\ \Delta_{ys} = \dfrac{2\pi}{B(k_{ys})} \end{cases} \tag{58}$$

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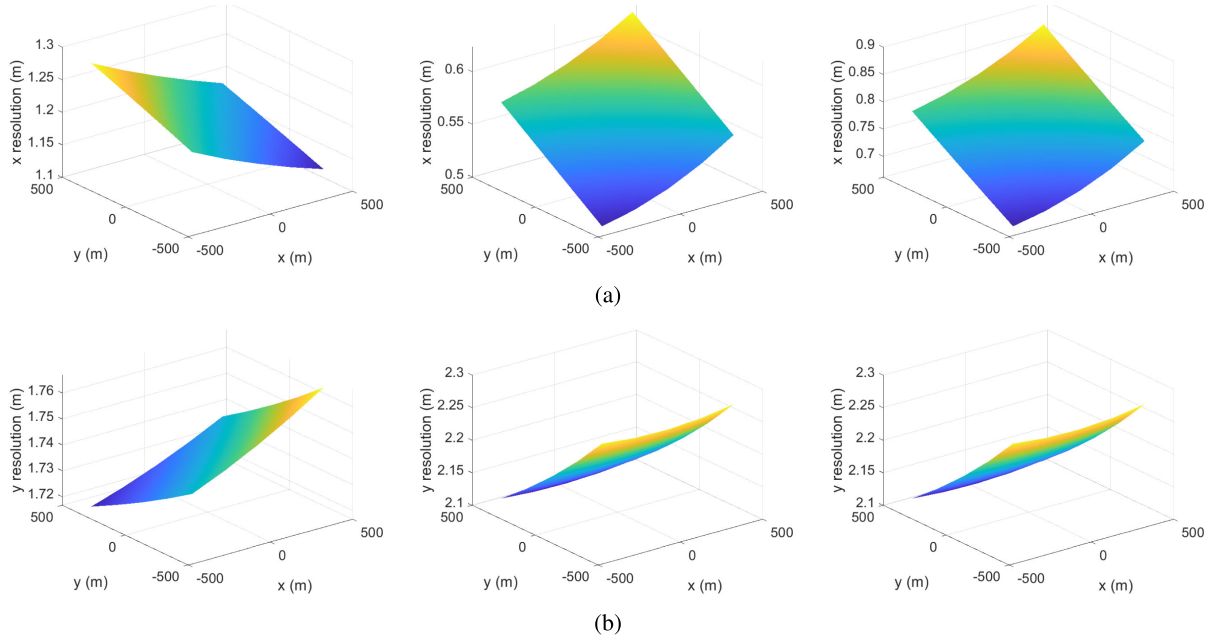


Fig. 8. Ground resolutions vary with Configurations I–III. (a) x resolution. (b) y resolution.

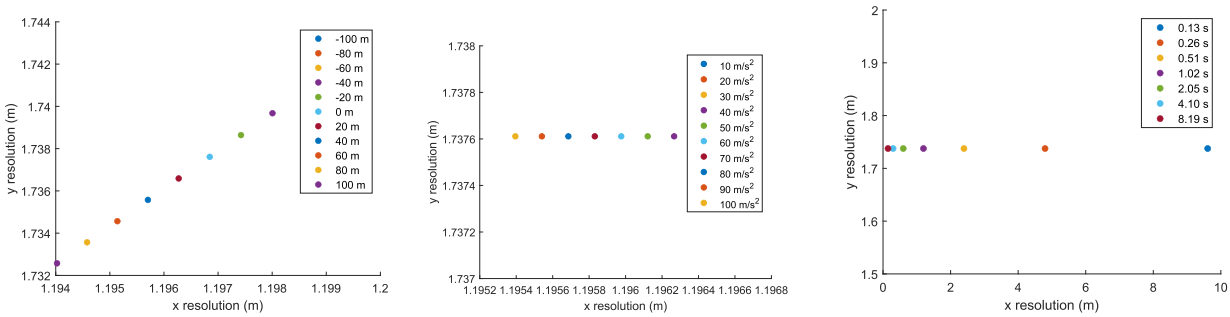


Fig. 9. Ground resolutions of the target at (100, 100, 0) in Configuration I vary with (a) z coordinate, (b) accelerations, and (c) synthetic time.

TABLE III
HIGH-ORDER MOTION PARAMETERS BEFORE ROTATION

Transmitter Acceleration (m/s^2)	(19,35,-1)
Receiver Acceleration (m/s^2)	(20,-150,-50)
Transmitter Jerk (m/s^3)	(10,9,3)
Receiver Jerk (m/s^3)	(5,20,-10)

TABLE IV
TARGET COORDINATES IN $OXYZ$ (m)

P1	P4	P7
(-500.5,-500.5,0.8)	(-0.3,-500.5,-54.2)	(500.0,-500.5,-1.2)
P2	P5	P8
(-500.5,-0.3,-17.0)	(-0.3,-0.3,9.7)	(500.0,-0.3,18.5)
P3	P6	P9
(-500.5,500.0,1.4)	(-0.3,500.0,68.0)	(500.0,500.0,2.6)

and $B(k_{xs}) = \max(k_{xs}) - \min(k_{xs})$.

The 2-D wavenumber spectrum in Cartesian coordinates is shown in Fig. 4. It is noted that the range and azimuth wavenumbers are both decomposed into x- and y-directions; thus, the sizes of these Cartesian coordinate wavenumber support regions are not proportional to the synthetic aperture time. To reduce the sampling rate in the y-direction, the above parallelogram-shaped 2-D spectrum is rotated by the BRA to have the azimuth direction parallel to the k_x-axis [38]. The BRA in (59) is calculated with the scene center target, and its value depends on the initial bistatic configuration at t_0

$$\phi_b = -\arctan \frac{B[k_{xc}(k_{ra}, t_0)]}{B\left[k_{yc}(k_{ra}, t_0)\right]}. \tag{59}$$

Assume that the 2-D spectrum in Fig. 4 corresponds to the Cartesian coordinate system of data recording before Fig. 1. The first step in the imaging algorithm is to calculate the BRA and rotate the initial coordinate system to the Cartesian coordinate system $OXYZ$. On the basis of the rotated spectrum in Fig. 5, the 2-D Cartesian resolution of each target is about

$$\begin{cases} \Delta_x = \dfrac{2\pi}{B[k_x(k_{rc}, t_m)]} \\ \Delta_y = \dfrac{2\pi}{B\left[k_y(k_{ra}, t_0)\right]}. \end{cases} \tag{60}$$

Note that the shape of the 2-D spectrum, as well as the BRA, is spatial-variant, so there are still residual BRAs for the other targets after the uniform rotation.

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TABLE V
IMAGING QUALITY EVALUATION IN x

Method	PSLR(dB)/ISLR(dB)/IRW(m)			
	P2	P3	P4	P5
PFA	-10.97/-8.61/1.42	-6.27/2.06/-	-11.07/-10.06/1.22	-12.27/-9.55/1.25
BPPFA	-12.48/-10.08/1.32	-12.35/-9.82/1.68	-5.38/-5.89/1.44	-12.16/-10.19/1.20
BPA	-13.23/-10.36/1.35	-13.38/-10.90/1.67	-12.82/10.49/1.20	-13.16/-10.64/1.19
SA-BPPFA	-13.19/-10.36/1.35	-13.34/-10.90/1.67	-12.57/-10.24/1.20	-13.12/-10.37/1.19

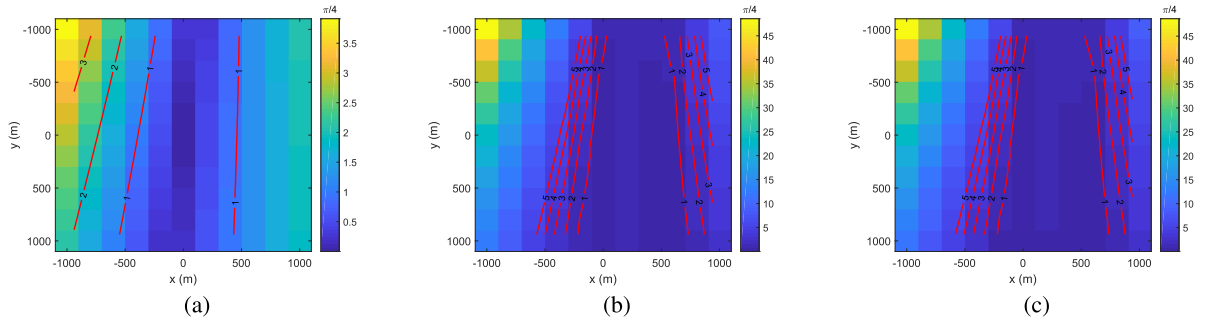


Fig. 10. Residual phase errors vary with different configurations. (a) I. (b) II. (c) III.

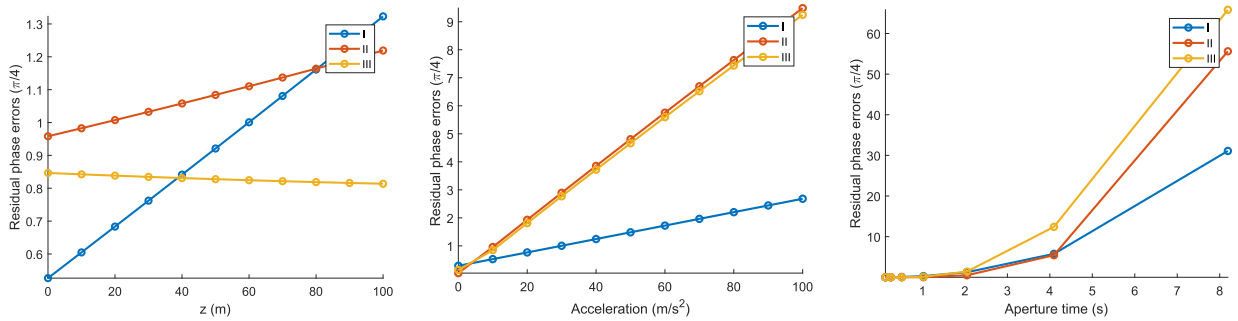


Fig. 11. Residual phase errors of the target at (500, 500, 0) vary with (a) z coordinate, (b) accelerations, and (c) synthetic time.

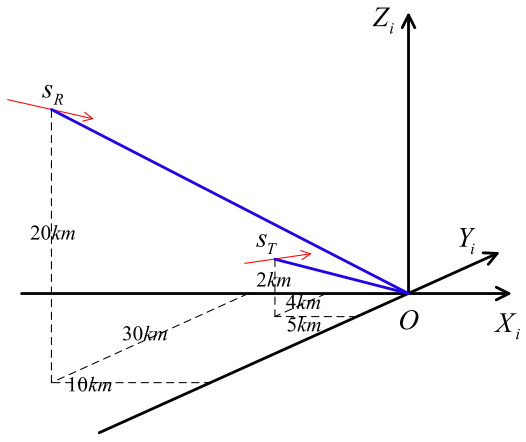


Fig. 12. Schematic of Configuration I in the initial coordinate system.

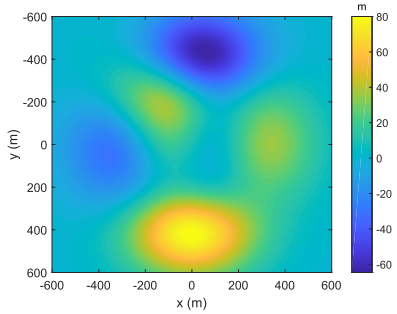


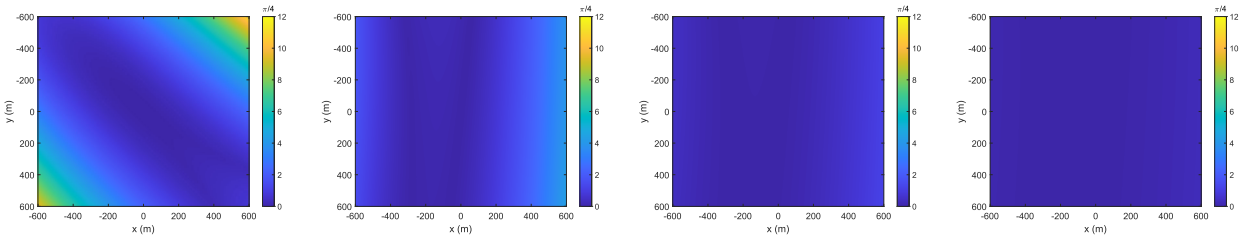
Fig. 13. Scene elevation.

C. Implementation of the SA-BPPFA

Sections III-A and III-B presented the details of coarse focus imaging with BPPFA and the Nyquist sampling requirements, and this section presents the flow of the proposed SA-BPPFA with the general SA processing embedded in it. The flowchart is shown in Fig. 6, and the proposed algorithm is summarized in the following steps.

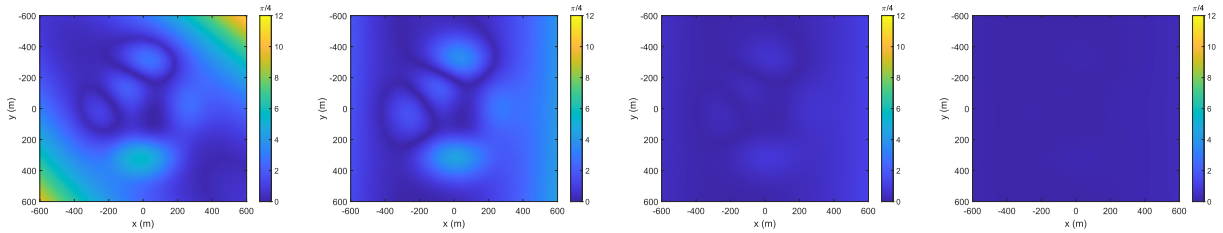
1) Starting from the right part of the flowchart, we first determine the bistatic configuration and the parametric coordinate; then, we calculate the BRA and rotate the initial data-recording Cartesian coordinate system to the $OXYZ$ coordinate system in Fig. 1 or 2.
2) With the help of the backgeocoded DEM and the focus requirements of resolution and DOF, the unified Cartesian Nyquist grids are prepared for back projection.

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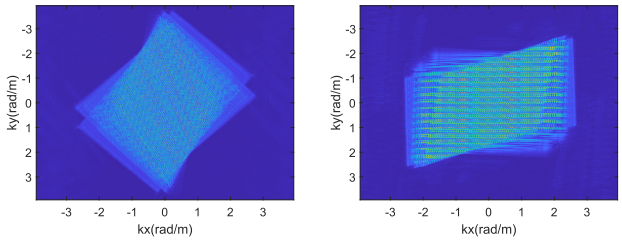
(a) (b) (c) (d)

Fig. 14. DOF of 0-m elevation scenario. (a) PFA. (b) BPPFA. (c) SA-BPPFA (two SAs). (d) SA-BPPFA (four SAs).



(a) (b) (c) (d)

Fig. 15. DOF of undulated scenario. (a) PFA. (b) BPPFA. (c) SA-BPPFA (two SAs). (d) SA-BPPFA (four SAs).



(a) (b)

Fig. 16. 2-D spectrum (a) before rotation and (b) after rotation.

3) Now, we come to the left part and split data as well as apertures to achieve coarse focus within SAs. Imitating (38), the coarse-focused image in the kth SA is shown in (61), at the bottom of the next page.
4) The schematic of the relationship between unified Cartesian coordinates and local SA polar coordinates is shown in Fig. 7. Since the coordinate transformation relation is configuration-dependent, combining (19) and (41), let $\mathbf{T}^{(k)}$ denote the imaging-Cartesian coordinate transformation in the kth SA and project the subimages into the unified Cartesian grids.
5) The phase compensation function for coherent addition between SAs is given as ($\tilde{R}(\varphi_s^{(k)}, \rho^{(k)}) = 0$)

$$H_{\text{ref}}^{(k)}\left(x_g, y_g\right) = \exp\left\{-jk_{rc}\left[\rho^{\text{all}}\left(x_g, y_g\right) - \rho^{(k)}\left(x_g, y_g\right)\right]\right\} \tag{62}$$

where (x_g, y_g) is the grid position of (x, y) and ρ^{all} is its bistatic range at the full aperture center. Then, the final fused image is

$$I\left(x_g, y_g\right) = \sum_{k=1}^{n} \mathbf{T}^{(k)}\left[I^{(k)}\left(\rho_s^{(k)}, \varphi_s^{(k)}; \rho^{(k)}, \varphi^{(k)}\right)\right] \cdot H_{\text{ref}}^{(k)}\left(x_g, y_g\right). \tag{63}$$

Now, determine whether the resolution requirements have been met, if not, collect new data and repeat steps 3)–5) until the final focused image is obtained.

IV. Practical Considerations

This section provides the necessary analysis of the practical considerations, including the focus performance and the computational complexity of the proposed SA-BPPFA.

A. Focus Performance Analysis

At first, we analyze the focusing performance in terms of both resolution and DOF. By summarizing the above derivations, the main factor affecting the DOF is the residual phase error after azimuth interpolation. However, the coupling of radar system parameters, motion parameters, and imaging requirements makes it difficult to quantify the limitation of a single parameter. Focusing on the focus performance, we qualitatively analyze the trends of ground resolution and DOF varying with configurations, 3-D positions, radar motion errors, and the synthesis duration. The radar system parameters are given in Table I, and three configuration parameters are given in Table II. Configuration I consists of a hypersonic dive receiver with a flat-track airborne transmitter; Configurations II and III, with the same initial configuration, are composed of two airborne platforms. Configuration II is the translational-invariant one along with motion errors, and Configuration III equips the stationary receiver. Configurations I and II (or III) are used to compare the effect of the initial configuration on performance, and Configurations II and III are used to compare the effect of the radar motion parameters on performance (these three configurations are set in the initial data-recording coordinate system $OX_iY_iZ_i$ before $OXYZ$).

1) Resolution: According to (60), Fig. 8 shows the ground x and y resolutions vary with configurations. Unlike the first

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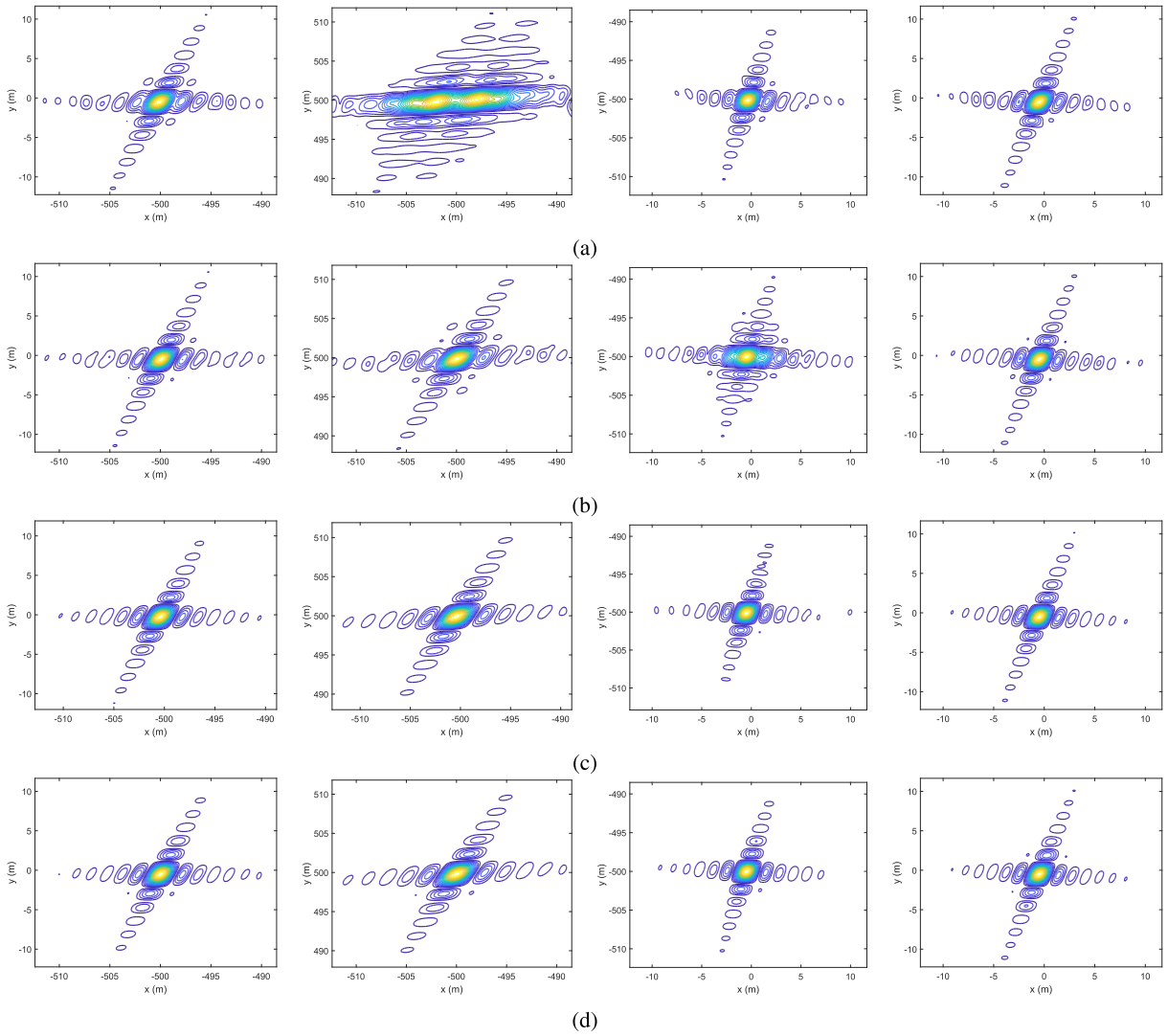


Fig. 17. Focus results of (a) PFA, (b) BPPFA, (c) BPA, and (d) SA-BPPFA (four SAs) (the targets from left to right are $P2$–$P5$).

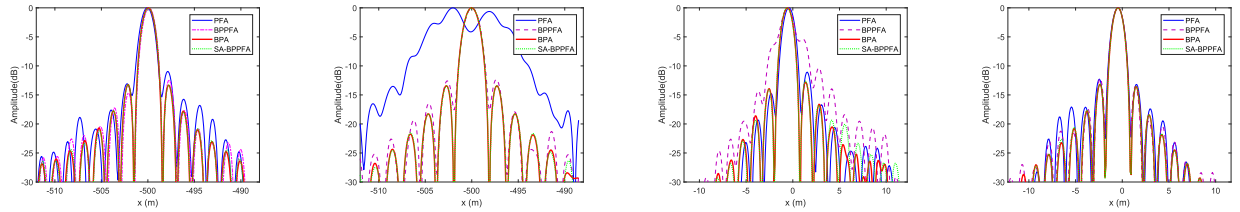
column, the second and third columns show the same trend, suggesting that the initial bistatic configuration affects the resolution distribution at the same scene location. A comparison of the second and third columns shows that their x resolutions are different, and the y resolutions are the same. Because the rotation of the bistatic spectrum results in the y resolution being derived from the range resolution of the original echo data, the x resolution is derived from the Doppler resolution of the echo data, so that the x resolution is higher in the case with a longer synthetic aperture. In Fig. 9(a) and (b), the ground resolutions slightly vary with targets' elevations and the radar's accelerations; in Fig. 9(c), it is further demonstrated that the x resolution is approximately proportional to the synthetic aperture length, while the y resolution is independent of it.

Overall, the main factors affecting the resolution of the BSSAR are the initial configuration and the synthetic aperture length. In these cases, the maximum difference of the spatial-variant resolution is about 0.25 m for this scenic area, and it is noted that the design of the bistatic configuration requires attention to the resolution requirements of the observation scenario.

2) DOF: According to (42), Fig. 10 shows the residual phase errors vary with configurations, and the DOF is generally defined as the area where the phase errors are less than $\pi/4$. In Fig. 10, the second and third columns have the same initial bistatic configurations and contour alignment, while the contours in Configuration I are sparser. The effective focus scene size is essentially the same for all three configurations, but the distribution position differs. In Fig. 11, the effects of

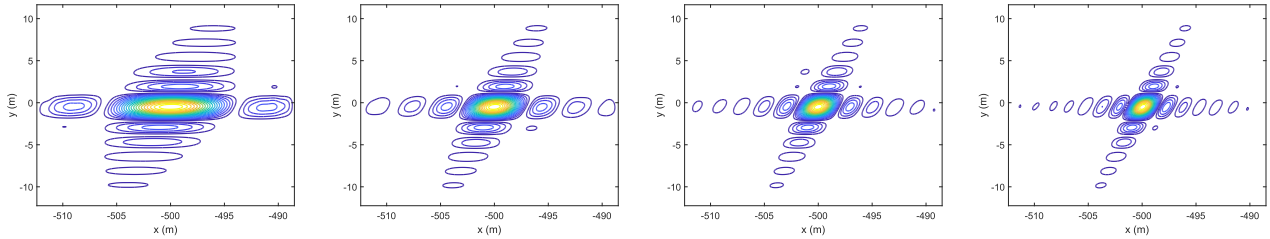
$$I^{(k)}\left(\rho_s^{(k)},\varphi_s^{(k)};\rho^{(k)},\varphi^{(k)}\right)=\varpi\left(\rho^{(k)}\right)\operatorname{sinc}\left(\frac{\rho_s^{(k)}-\rho^{(k)}}{\Delta_\rho^{(k)}}\right)\sin c\left[\frac{\varphi_s^{(k)}-\psi\left(\rho^{(k)},\varphi^{(k)},z\right)}{\Delta_\varphi^{(k)}}\right]\exp\left\{-jk_{rc}\left[\rho_s^{(k)}+\tilde{R}\left(\varphi_s^{(k)},\rho_s^{(k)}\right)\right]\right\} \tag{61}$$

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(a) (b) (c) (d)

Fig. 18. x-direction profiles of (a) $P2$, (b) $P3$, (c) $P4$, and (d) $P5$.



(a) (b) (c) (d)

Fig. 19. Number of fused SAs is (a) 1, (b) 2, (c) 3, and (d) 4.

target elevation, acceleration, and synthetic time on the residual phase error are compared. It indicates that the DOF varies most dramatically with the target elevation in Configuration I, and the elevation effect is reversed in Configurations II and III. Besides, the acceleration and the large duration have more influence on the low-speed platforms. Furthermore, combined with Fig. 9(c), the increase in resolution will be accompanied by a sharp reduction of DOF.

In summary, the bistatic configuration, the spatial distribution of the target, the track motion error, and the synthetic aperture length all have a significant impact on the DOF of the algorithm.

B. Computational Complexity

In this section, we calculate the computational complexity of the proposed algorithm. As is known, an FT or IFT of length N requires $5N\log_2 N$ floating-point operations (FLOPs), and a complex phase multiplication demands six FLOPs. When the interpolator length is k_{ker}, one interpolation needs $N_{\text{ker}} = 2(2k_{\text{ker}} - 1)$ FLOPs [36]. We assume the number of azimuth pulses, the number of range samples, the number of SAs, the number of the grids of the x-direction, the number of the grids of the y-direction, and the number of the grids of DEM are N_a, N_r, n, N_x, N_y, and N_{DEM}, respectively.

The computational complexity of the proposed algorithm is divided into three parts: coarse focus with BPPFA, backgeocoding DEM interpolation, and subimage fusion. The computation of the monostatic PPFA is analyzed in [34]; thus, the computational burden of coarse focus is

$$O_1 = N_r N_a \cdot \left[24 + N_{\text{ker}} + 10\log_2(N_r) + 20\log_2(N_a) - 20\log_2(n)\right]. \quad (64)$$

The computational burden of backgeocoding DEM interpolation is

$$O_2 = N_x N_y N_{\text{DEM}} N_{\text{ker}}. \quad (65)$$

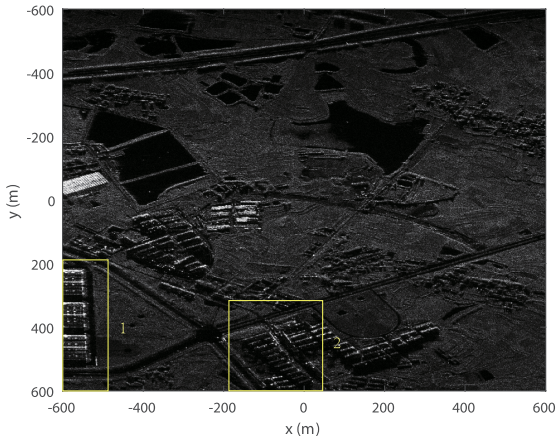


Fig. 20. Scene scattering coefficient from a monostatic SAR image.

The subimage fusion includes signal interpolation and complex phase multiplication, and the computational burden of subimage projection is

$$O_3 = N_x N_y N_r N_a (6 + N_{\text{ker}}). \quad (66)$$

So, the total computational burden of SA-BPPFA is

$$O_{\text{total}} = O_1 + O_2 + O_3. \quad (67)$$

It can be seen that the more the SAs, the higher the efficiency.

In SA focusing, SA-BPPFA employs the modified PFA. For SA fusion, the subimages are directly projected to the unified Cartesian coordinate system, which is more efficient, because it only needs to perform the backgeocoding interpolation once. In contrast, BFFBPA requires the backgeocoding interpolation of the external DEM for each coordinate projection, which increases both the interpolation error and the number of operations. For comparison, we consider a bistatic fast BPA (BFBPA), which is degraded from BFFBPA, and the accumulation of interpolation errors is neglected. But, the BFBPA subimages also require backgeocoding interpolation in every projection. The computational burden of coarse focus for

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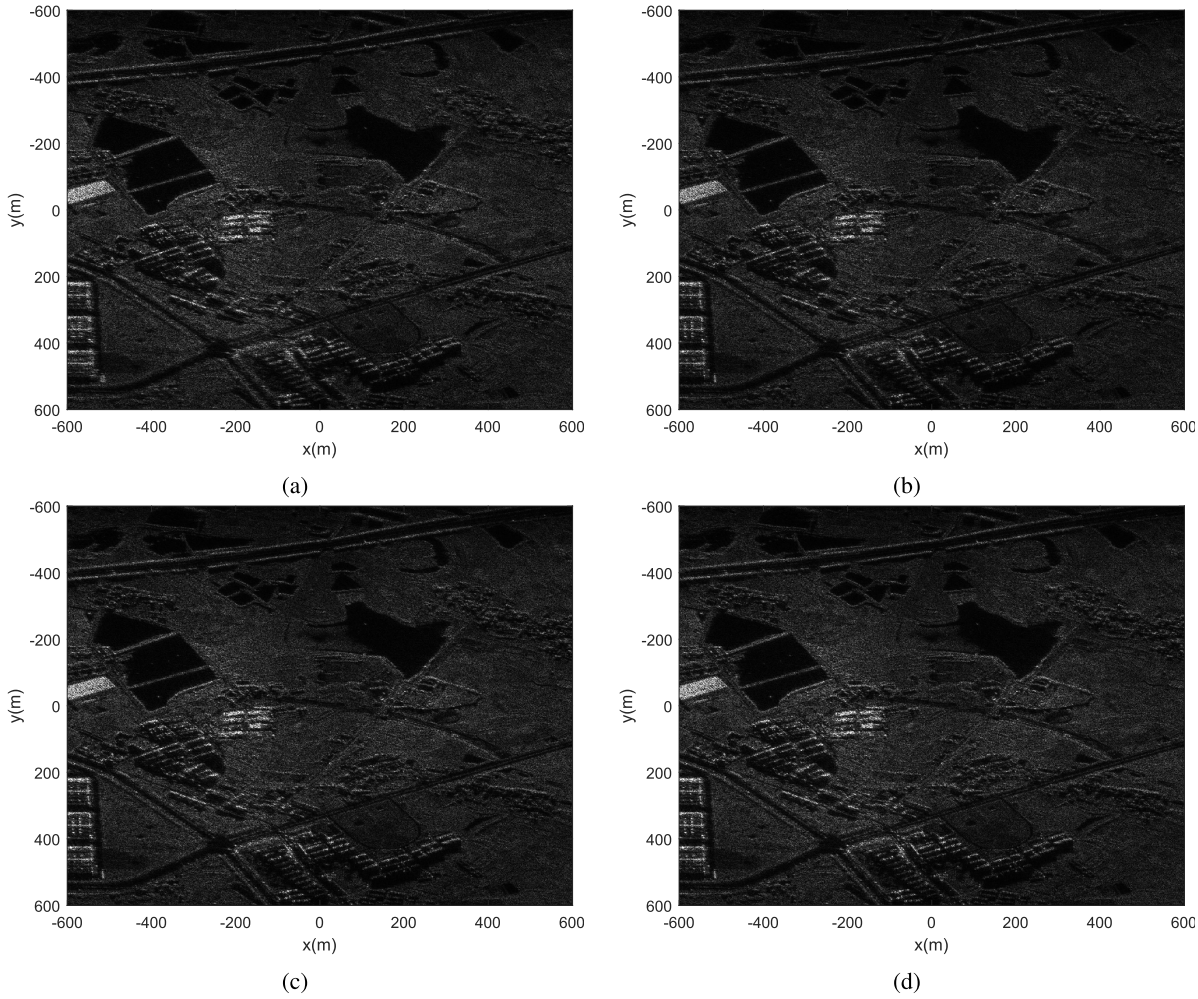


Fig. 21. Processing results of distributed target by (a) PFA, (b) BPPFA, (c) BPA, and (d) SA-BPPFA (four SAs).

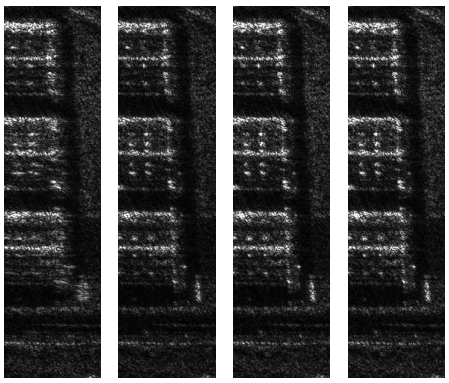


Fig. 22. Enlarged part 1 of Fig. 21 processed by (a) PFA, (b) BPPFA, (c) BPA, and (d) SA-BPPFA.

BFBPA is (assumed that the number of local polar coordinate grids for BFBPA is also $N_r(N_a/n)$)

$$O_1' = N_r {N_a}^2 (6 + N_{\text{ker}})/n. \tag{68}$$

The computational burden of backgeocoding DEM interpolation for BFBPA is

$$O_2' = \left(N_x N_y + N_r N_a\right) N_{\text{DEM}} N_{\text{ker}}. \tag{69}$$

The total computational burden of BFBPA is

$$O'_{\text{total}} = O_1' + O_2' + O_3. \tag{70}$$

In general, $O'_{\text{total}} \geq O_{\text{total}}$. Also, it is noted that, with the proper size of the Cartesian grid, the proposed algorithm enables fast SA imaging and SA fusion with data recording for real-time resolution enhancement. It is also clear that if a fast and high-precision interpolator is available, the proposed BSSAR algorithm can be extended to a fast-factorized framework.

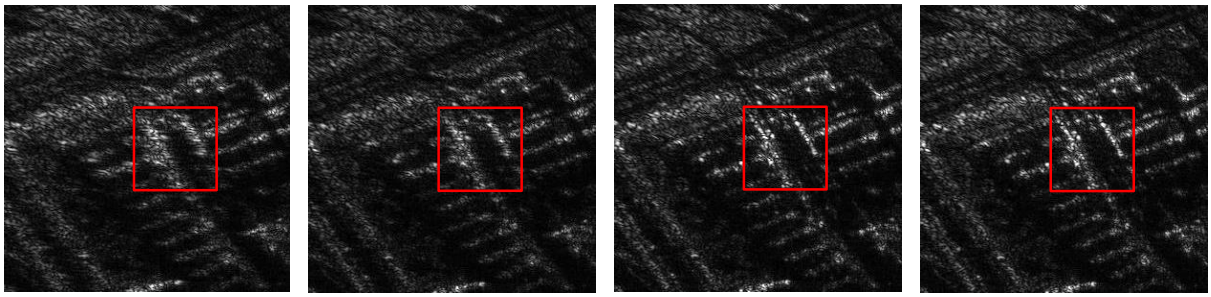
V. Simulation Results

A. Point Targets Simulation

In this section, the simulation of the point target is carried out with Configuration I in Table II, which is illustrated as Fig. 12, and the high-order motion parameters are displayed in Table III. Based on Fig. 10, the imaging size is 1200 × 1200 m ($x \times y$), and its elevation is given in Fig. 13, with a span of −64.7 to 80 m. The minimum resolution for this scenario is about 1 × 1 m according to Fig. 8, and the grid size of the final fused Cartesian coordinate is set as 0.8 × 0.8 m.

The effect of individual factors was analyzed in Section IV-A according to (42), whereas in the case of combined factors, the residual phase error of the proposed

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(a) (b) (c) (d)

Fig. 23. Enlarged part 2 of Fig. 21 processed by (a) PFA, (b) BPPFA, (c) BPA, and (d) SA-BPPFA.

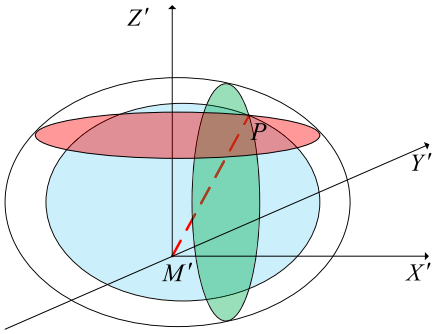


Fig. 24. Ellipsoidal coordinate system.

algorithm for different SA numbers is shown in Figs. 14 and 15. The results of the conventional PFA in [39] are provided, too. In Fig. 14, the scene is set on the zero elevation plane. For the conventional PFA, the residual phase errors increase with increasing x and y coordinates, and the effective DOF is limited in two dimensions, whereas the residual phase errors of SA-BPPFA mainly vary with the x coordinates. The maximum residual phase error for the whole scene is about 11 times, 4 times, 1.2 times, and 0.3 times $\pi/4$ in Fig. 14(a)–(d) for the number of SAs of 1, 1, 2, and 4, respectively. In Fig. 15, we add the elevations in Fig. 13. Obviously, the larger the elevation, the larger the residual phase errors. The maximum residual phase error caused by elevation is about 5.5 times, 4.5 times, 0.9 times, and 0.2 times $\pi/4$ in Fig. 15(a)–(d) for the number of SAs of 1, 1, 2, and 4, respectively. The analysis of Figs. 14 and 15 shows that, such as PFA, the residual phase error is also spatial-variant for our algorithm, which limits its DOF in three dimensions. However, by combining SAs, the effective DOF of the proposed algorithm is extended, and the more the SAs, the greater the DOF. But, the more the SAs, the greater the interpolation accumulation error, and the minimum number of SAs to ensure a fully effective focus on this experimental terrain is 4.

Next, we select nine point targets from the scene map (Fig. 13) and give their focus results. For the nine point targets in Table IV, the 2-D spectra before and after rotation according to the BRA are shown in Fig. 16 (the spatial-variant wavenumber centers have been aligned). As analyzed in Section III-B, the spectral rotation reduces the requirement for sampling in the y-direction, while ensuring that the x-direction point response function is approximately parallel to the coordinate axis.

TABLE VI
CONTRAST AND ENTROPY COMPARISON

Methods	BPA	PFA	BPPFA	SA-BPPFA(4SAs)
Contrast	0.451	0.447	0.447	0.449
Entropy	13.48	13.50	13.50	13.49

Fig. 17 presents contour plots for $P2$–$P5$, which are processed by SA-BPPFA, PFA in [39], and BPA (the SA BPA strategy in the literature [26] is limited by the radar track model, and it fails completely without additional modifications). To better illustrate the role of SAs, Fig. 17 provides the imaging results of the one-SA (full aperture) and four-SAs cases, respectively. Fig. 18 also displays the x profiles, and Table V gives the quantitative results in terms of the peak sidelobe ratio (PSLR), the integrated sidelobe ratio (ISLR), and the impulse response width (IRW) of the targets in Fig. 17.

The four targets represent the x-direction edge target with a large elevation, the 2-D edge target with a small elevation, the y-direction edge target with a much larger elevation, and the scene center target with a large elevation, respectively. Corresponding to Fig. 15, the azimuth defocusing occurs in the PFA results of targets $P2$–$P4$, and as the 2-D edge target, the result of $P3$ is the worst, whereas the results of $P2$ and $P4$ have roughly the same imaging quality referring to Table V and Fig. 18. BPPFA (the one-SA BPPFA) has slightly better performance than PFA, with the worst imaging quality being target $P4$, while $P2$ and $P3$ are roughly the same. When the number of SAs is 4, all four results of SA-BPPFA are well focused, and their focusing quality is comparable to that of the BPA, which is also evidenced in Table V and Fig. 18.

In order to show the resolution improvement caused by the SA fusion, the contour maps of $P2$ with different numbers of fused aperture are given in Fig. 19 (the total number of SAs is 4).

In summary, the bistatic PFA has a limited 3-D DOF at the scene edge and high elevation. However, by increasing the number of SAs, the SA-BPPFA can continuously improve the DOF while maintaining resolution. The final focus performance of a four-SAs BPPFA is similar to that of the BPA.

B. Distributed Scene Simulation

In this part, the simulation on a distributed target scene is carried out at last. The simulation parameters and the scene settings keep consistent with the previous point target

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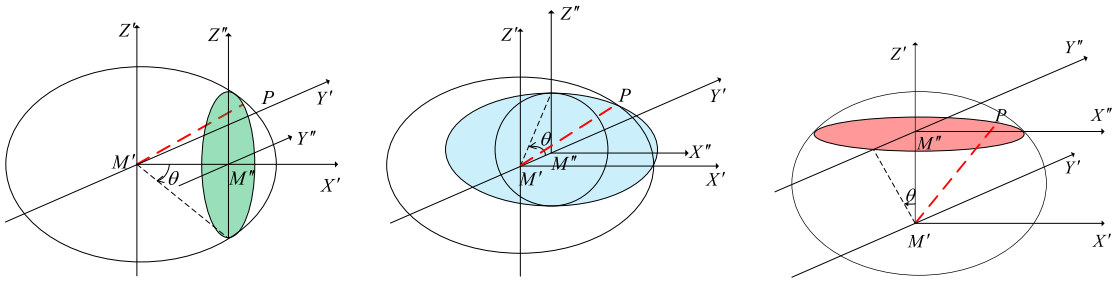


Fig. 25. Different projections onto (a) $M''Y''Z''$ plane, (b) $M''X''Z''$ plane, and (c) $M''X''Y''$ plane.

simulation. Only to verify the focusing performance of the algorithm, a monostatic SAR image in Fig. 20 is used as the ground-truth map, and the pixel size is 1.2 × 1.2 m. The BSSAR echo is calculated pulse by pulse and target by target according to Configuration I in Table II [40].

Similar to the point target simulation, the imaging results and the local parts obtained by the four algorithms are shown in Figs. 21–23. To quantitatively evaluate the performance of the algorithm, the image entropy and image contrast in Table VI are calculated for the image results in Fig. 21. Consistent with the point target simulation analysis, bottom-left part 1 and more undulating part 2 of the PFA results are both defocused in Figs. 22(a) and 23(a). For the full-aperture BPPFA, its imaging result of part 2 in Fig. 23(b) also defocuses seriously, whereas the results of the proposed SA algorithm are close to those of the BPA in Figs. 21–23, verifying the imaging effectiveness of SA-BPPFA for complete large scenes. The values in Table VI also confirm these conclusions. Table VII gives the running time varies with the scene grids, and it is noted that increasing the number of grids will inevitably increase the processing time of all the algorithms, whereas increasing the number of SAs reduces the running time of the proposed algorithm. SA-BPPFA takes about the same amount of time as PFA, and the more the samples in the scene, the more the efficient it is than BPA.

In summary, the proposed algorithm has comparable processing efficiency and the ability to continuously extend the DOF compared with PFA, and comparable focus quality and higher efficiency compared with BPA.

TABLE VII
RUNNING TIME (RM S)

Scene grids	BPA	PFA	BPPFA	SA-BPPFA(4SAs)
100 × 100	2.3	6.3	7.1	2.7
500 × 500	27.8	7.3	9.7	4.5
1500 × 1500	309.0	21.3	35.0	25.0

VI. CONCLUSION

This article proposes an SA PFA built on the coordinate-transformation-based polar coordinate system and the DSR slant range model. By applying the ellipsoid parametric equation, BPPFA provides the motion-compensated focused results and the coordinate transformation relationship first, relying on which the SA images are fused regardless of trajectory and configuration. In turn, the reduction of aperture length enables SA-BPPFA to improve the DOF and azimuth resolution of BPPFA concurrently. On the other hand, with comparable image quality, SA-BPPFA is much more efficient than the most accurate BPA. All the simulations verify our algorithm's imaging capability and algorithmic superiority for high-resolution and complex-trajectory BSSAR with general configuration.

APPENDIX

A. Ellipsoidal Parametric Equations

$$\begin{cases} x' = a\cos\theta \\ y' = b\sin\theta\cos\varphi \\ z' = b\sin\theta\sin\varphi \end{cases}, \quad 0 \le \theta \le \pi \,\&\, 0 \le \varphi < 2\pi \tag{71}$$

$$\begin{cases} x' = a\sin\theta\cos\varphi \\ y' = b\cos\theta \\ z' = b\sin\theta\sin\varphi \end{cases}, \quad 0 \le \theta \le \pi \,\&\, 0 \le \varphi < 2\pi \tag{72}$$

$$\begin{cases} x' = a\sin\theta\cos\varphi \\ y' = b\sin\theta\sin\varphi \\ z' = b\cos\theta \end{cases}, \quad 0 \le \theta \le \pi \,\&\, 0 \le \varphi < 2\pi. \tag{73}$$

$$\begin{aligned} n_x &= \frac{\partial x}{\partial \varphi} = a\cos\alpha\cos\beta(-\sin\varphi\sin\theta + n_s\cos\varphi) - b\sin\alpha n_c - b\cos\alpha\sin\beta(\cos\varphi\sin\theta + n_s\sin\varphi) \\ n_y &= \frac{\partial y}{\partial \varphi} = a\sin\alpha\cos\beta(-\sin\varphi\sin\theta + n_s\cos\varphi) + b\cos\alpha n_c - b\sin\alpha\sin\beta(\cos\varphi\sin\theta + n_s\sin\varphi) \\ n_s &= \frac{\partial\sin\theta}{\partial\varphi} = \frac{1}{z_M - z_c}(-a\sin\beta\sin\varphi + b\cos\beta\cos\varphi)\sin^2\theta \\ n_c &= \frac{\partial\cos\theta}{\partial\varphi} = -\tan\theta\cdot\frac{\partial\sin\theta}{\partial\varphi} \end{aligned} \tag{75}$$

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Fig. 24 shows the ellipsoidal coordinate system $M'X'Y'Z'$, and three parametric equations (71)–(73) correspond to the subgraphs of Fig. 25, respectively.

B. Derivations of the Scaling Azimuth Time and Its Derivative

$$\begin{aligned} f_x &= \frac{\partial R(t_m; x, y, z_c)}{\partial x} \\ &= -\frac{x_r + \Delta_{rx}(t_m) - x}{R_r(t_m; x, y, z_c)} - \frac{x_t + \Delta_{tx}(t_m) - x}{R_t(t_m; x, y, z_c)} \\ f_y &= \frac{\partial R(t_m; x, y, z_c)}{\partial y} \\ &= -\frac{y_r + \Delta_{ry}(t_m) - y}{R_r(t_m; x, y, z_c)} - \frac{y_t + \Delta_{ty}(t_m) - y}{R_t(t_m; x, y, z_c)} \end{aligned} \tag{74}$$

and n_x and n_y are shown in (75), at the bottom of the previous page; then

$$\begin{aligned} & g_r(t_m; \rho, \varphi, z_c) \\ &\quad = g_{r1} + g_{r2} \\ & g_{r1} = -\frac{\mathbf{n} \cdot \frac{\partial \Delta_r(t_m)}{\partial t_m}}{R_r(t_m; x, y, z_c)} \\ & g_{r2} = \frac{\left[\hat{R}_r(t_m; x, y, z_c) \cdot \mathbf{n}\right]\left[\hat{R}_r(t_m; x, y, z_c) \cdot \frac{\partial \Delta_r(t_m)}{\partial t_m}\right]}{R_r(t_m; x, y, z_c)} \end{aligned} \tag{76}$$

$$\begin{aligned} & g_t(t_m; \rho, \varphi, z_c) \\ &\quad = g_{t1} + g_{t2} \\ & g_{t1} = -\frac{\mathbf{n} \cdot \frac{\partial \Delta_t(t_m)}{\partial t_m}}{R_r(t_m; x, y, z_c)} \\ & g_{t2} = \frac{\left[\hat{R}_t(t_m; x, y, z_c) \cdot \mathbf{n}\right]\left[\hat{R}_t(t_m; x, y, z_c) \cdot \frac{\partial \Delta_t(t_m)}{\partial t_m}\right]}{R_t(t_m; x, y, z_c)} \end{aligned} \tag{77}$$

and

$$\begin{aligned} \mathbf{n} &= \left[n_x, n_y, 0\right] \\ \hat{R}_r(t_m; x, y, z_c) &= \frac{\mathbf{R}_r(t_m; x, y, z_c)}{R_r(t_m; x, y, z_c)} \\ \hat{R}_t(t_m; x, y, z_c) &= \frac{\mathbf{R}_t(t_m; x, y, z_c)}{R_t(t_m; x, y, z_c)} \\ \Delta_r(t_m) &= \left[\Delta_{rx}(t_m), \Delta_{ry}(t_m), \Delta_{rz}(t_m)\right] \\ \Delta_t(t_m) &= \left[\Delta_{tx}(t_m), \Delta_{ty}(t_m), \Delta_{tz}(t_m)\right]. \end{aligned} \tag{78}$$

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